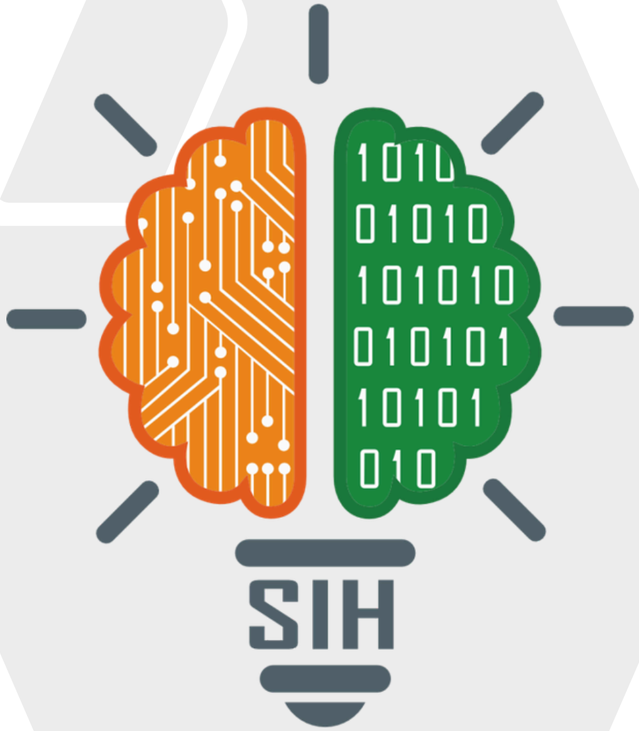
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TITLE PAGE

- **Problem Statement ID –** 26146
- **Problem Statement Title-** AI-Powered Monitoring & Analysis of Bitcoin Transaction Traffic
- **Theme -** Blockchain & Cybersecurity
- **PS Category-** Software
- **Team ID-**
- **Team Name -** Lazarus



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CHAKRAVYUH

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Our Solution

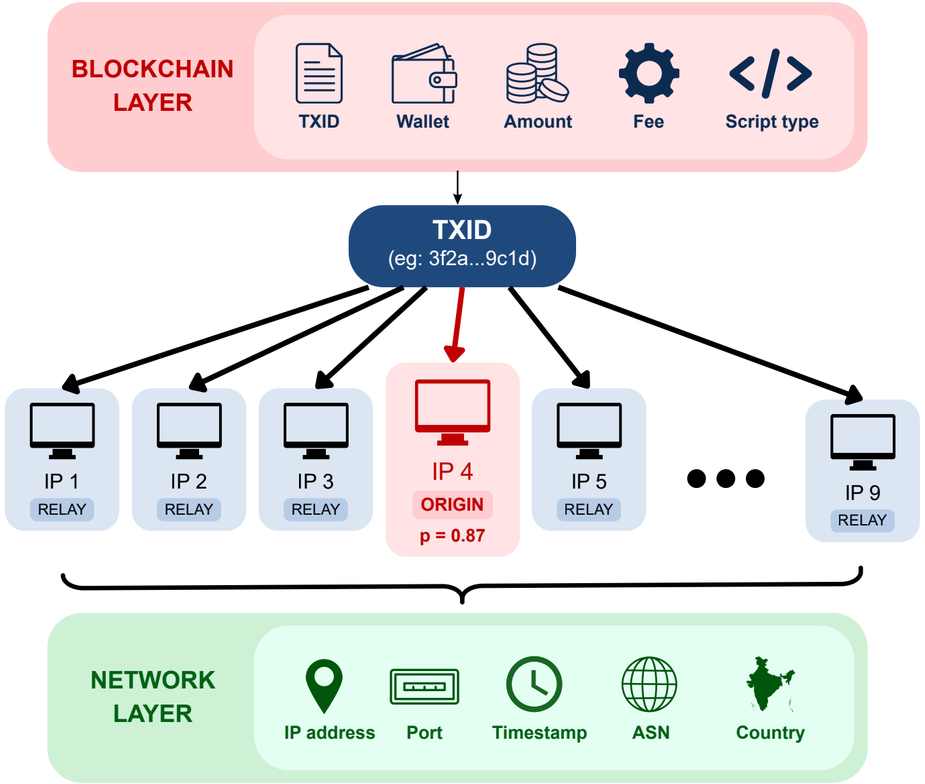
STEP 01 — Input
Bulk CSV/ JSON/ XML
(network + blockchain)

STEP 02 — Two-Layer Fusion
Builds a dual graph combining blockchain and network data

STEP 03 — AI/ML Analysis
- Address clustering
- Anomaly detection
- Peel-chain / mixing detection
- Taint propagation

STEP 04 — Output
Ranked alerts with confidence
Signed attribution packet (PDF + JSON)

Same TXID, nine announcing peers.
Only one may be the originator.



Innovation & Uniqueness

Origin VS Relay Estimator
Scores every announcing peer, resolves the shortlist to operator (ASN + region), and keeps the exact host sealed

Calibrated Confidence
Class-conditional conformal prediction

Open Benchmark
Synthetic dataset with ground truth

Counter-Evidence
Shows strongest case against each flag

How it addresses the problem

Bridges the Structural Gap
Chain analytics cannot see who transmitted. Traffic analysis cannot see what was moved. CHAKRAVYUH joins both layers.

Turns Data Into Leads
10^5 capture rows → 20 ranked leads that a three-officer cyber cell can review in one shift

Makes Every Lead Defensible
Each flag stores its cause, confidence and counter-argument, so investigators can defend or discard it

Keeps Sensitive Data Offline
All interception metadata stays on the air-gapped workstation and never leaves the system

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TECHNICAL APPROACH

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AIR-GAPPED HOST (offline)

MAYAJAAL
Gossip Network Simulator
Simulates P2P transaction propagation with multi-observer announcements, known origin truth, and adversarial laundering behaviours

KAVACH
Intake & Seal
Secures incoming artifacts with hashes and maintains the custody trail.

SETU
Normalize & Enrich
Standardizes timestamps and adds geographic and network context.

JAAL
Fused Graph
Connects transactions, addresses, IPs, ASNs, countries, wallets and entities.

SHASTRA
Forensic Heuristics
Applies clustering, peeling, mixing and origin-vs-relay analysis.

BUDDHI
Models
LightGBM (baseline)
Conformal confidence
Risk score + confidence

VAANI
Explanation
Turns model decisions into evidence-grounded, human-readable explanations.

DRISHTI
Console
Presents ranked suspects, transaction links and origin locations for investigation.

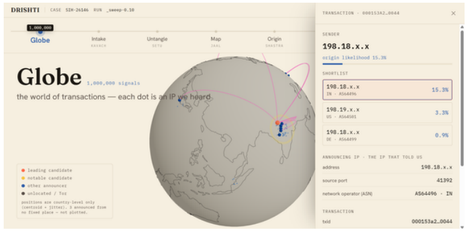
PRAMAAN
Attribution Packet
Packages signed evidence, hashes and the final legal notice for submission.

Phase 0 · Phase 1 · Phase 2 · Phase 3 · Phase 4 · Phase 5 · Phase 6 · Phase 7 · Phase 8

TECH STACK

- **Backend**: python · Polars · Apache Arrow · rustworkx · FastAPI
- **Frontend**: react · TypeScript · vite · three.js · lenis · gsap
- **ML**: LightGBM · PyTorch geometric · MAPIE · Shap · SciPy
- **Security**: SHA 256 file hashing · Merkle log (append-only log) · Ed25519 Digital signatures · ReportLab
- **Enrichment**: docker · archlinux · MAXMIND

Prototype wireframe







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FEASIBILITY AND VIABILITY

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Feasibility

01 **Runs on one laptop**
16 GB RAM, 8 cores, ~50 GB disk. GPU optional for GNN training and WebGL visualization.

02 **Processes millions of rows locally**
Polars handle millions of rows in-process, with no cluster or cloud required.

03 **Truly Offline**
Open-source dependencies are vendored into the image. Network adapter remains disabled.

04 **8-Layer Gated Plan**
Hard development gates ensure each completed stage runs end-to-end, even if later stages slip

RISK	WHAT WE DO
Dataset Link: Nil	MAYAJAAL: Simulated gossip network with known originator ground truth.
Entity Over-Merging	Never hard-merge clusters. Use confidence-weighted SAME_OWNER edges and expose the heuristic behind each link
Origin Estimation Defeated	Test against Tor, VPN and randomized relay delays. Report limits and abstain when evidence is insufficient.
Severe Class Imbalance	Use class-conditional conformal calibration with an explicit abstain band.
LightGBM may miss graph structure	Add graph-derived features such as degree, motifs and flow paths to LightGBM. Optional GNN embeddings are benchmarked separately
Innocent Nodes Flagged	Entity typing before scoring, mandatory counter-explanation, and human review before action.

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IMPACT AND BENEFITS

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Impact

01 **Makes Crypto Crime Traceable**
Illicit funds increasingly move through crypto, where investigations often hit dead end. Origin-focused tracing keeps cases moving

02 **Targets a Real, Growing Loss**
India lost ₹19,813 crore to fraud in 2025 (I4C, NCRP). Tracing suspicious crypto flows supports asset recovery and prosecution

03 **Turns Wallets Into Leads**
A wallet address alone does not reveal a person's identity. Linking it with IP, port and timestamp data can provide investigators with actionable leads

04 **Protects Sensitive Case Data**
Offline, agency-controlled deployment keeps sensitive investigation data within government infrastructure instead of relying on external cloud services

05 **Strengthens Investigation Decisions**
Complex blockchain activity can be converted into clear, actionable intelligence, helping investigators focus on the leads that matter most

Benefits

01 **Faster Triage**
Quickly scans large transaction volumes and highlights the wallets and patterns that need attention.

02 **Connected Evidence**
Links blockchain activity with network data to build stronger, more complete cases

03 **Explainable Leads**
Shows why a wallet or transaction was flagged, giving investigators clear evidence behind every lead.

04 **Fewer False Leads**
Separates strong evidence from weak signals and highlights counter-evidence to avoid wrong conclusions.

05 **Secure & Verifiable**
Works offline in an air-gapped environment, with every step securely logged and verifiable.

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RESEARCH AND REFERENCES

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Research

Deanonymisation of Clients in Bitcoin P2P Network
https://arxiv.org/abs/1405.7418

Transaction Clustering Using Network Traffic Analysis for Bitcoin and Derived Blockchains
https://orbilu.uni.lu/handle/10993/39728

Dandelion++: Lightweight Cryptocurrency Networking with Formal Anonymity Guarantees
https://arxiv.org/abs/1805.11060

On the Anonymity of Peer-to-Peer Network Anonymity Schemes Used by Cryptocurrencies
https://arxiv.org/abs/2201.11860

How Reliable Is the Multi-Input Heuristic for Bitcoin Address Clustering in Law Enforcement Contexts?
https://arxiv.org/abs/2607.07414

Resurrecting Address Clustering in Bitcoin
https://arxiv.org/abs/2107.05749

A Fistful of Bitcoins: Characterizing Payments Among Men with No Names
https://dl.acm.org/doi/10.1145/2504730.2504747

Anti-Money Laundering in Bitcoin: Experimenting with Graph Convolutional Networks for Financial Forensics
https://arxiv.org/abs/1908.02591

Elliptic++ (2306.06108)
https://arxiv.org/abs/2306.06108

Critical Re-Evaluation of GNNs under Temporal Shift (2604.19514)
https://arxiv.org/abs/2604.19514

A Gentle Introduction to Conformal Prediction (2107.07511)
https://arxiv.org/abs/2107.07511

Safeguarding the Evidential Value of Forensic Cryptocurrency Investigations
https://arxiv.org/abs/1906.12221

Reference

I4C / NCRP - Cyber Fraud Losses in India, 2025
https://www.prameyanews.com

Prototype

Github repo link: https://github.com/Veer-Raghava/chakravyuh
Prototype link:
Video link: